

Menstrual Disorders & Hormone Therapy in Women

Based on Pharmacotherapy Handbook, 7th Edition

Prepared for Pharmacy Yr IV UG Students



Edit with WPS Office

Menstrual Disorders – Overview

- Menstrual disorders include amenorrhea, dysmenorrhea, menorrhagia, and abnormal uterine bleeding.
- Associated with hormonal imbalance, structural causes, or systemic disease.
- Management tailored based on underlying etiology.



Definitions & Symptoms

Feature	Dysmenorrhea	Menorrhagia	Abnormal Uterine Bleeding (AUB)
Definition	Painful menstruation with cramping pelvic pain during menses	Excessive menstrual bleeding (>80 mL or >7 days) at regular intervals	Any bleeding abnormal in volume, regularity, or timing, not related to pregnancy
Key Symptoms	Pelvic pain, cramps, nausea, headache, fatigue	Heavy flow, clots, prolonged periods, anemia symptoms	Irregular cycles, unpredictable bleeding, spotting, prolonged/shortened cycles



Etiology & Clinical Impact

Feature	Dysmenorrhea	Menorrhagia	Abnormal Uterine Bleeding (AUB)
Etiology	Primary: prostaglandin-mediated contractions; Secondary: endometriosis, fibroids, PID	Structural (fibroids, adenomyosis), coagulopathies, hormonal imbalance	PALM-COEIN causes (Polyp, Adenomyosis, Leiomyoma, Malignancy; Coagulopathy, Ovulatory dysfunction, Endometrial, Iatrogenic, Not classified)
Clinical Impact	Reduced quality of life, absenteeism, psychological distress	Iron-deficiency anemia, fatigue, impaired functioning	Diagnostic challenge, anemia risk, may mask malignancy



Diagnosis & Management

Feature	Dysmenorrhea	Menorrhagia	Abnormal Uterine Bleeding (AUB)
Diagnosis	Clinical history, pelvic exam, exclude secondary causes	Menstrual history, CBC, ultrasound, hysteroscopy	Systematic evaluation (PALM-COEIN), labs, imaging, biopsy if indicated
Management	NSAIDs, hormonal contraceptives, lifestyle measures; treat underlying cause	Iron therapy, hormonal options (OCPs, LNG-IUS), surgical (ablation, hysterectomy)	Etiology-based: medical (hormones, antifibrinolytics), surgical (myomectomy, hysterectomy), supportive care
Resource-Limited Settings	Affordable NSAIDs, counseling, screen for secondary causes	Iron therapy, simple hormonal options, clinical diagnosis when imaging	Clinical classification, rule out pregnancy/malignancy, pragmatic labs

Management of Menstrual Disorders (Updated ACOG 2024)

American College of Obstetricians and Gynecologists

- **Dysmenorrhea**
 - **First-line:** NSAIDs (ibuprofen, naproxen) to reduce prostaglandin-mediated pain.
 - **Hormonal therapy:** Combined oral contraceptives, progestin-only methods, or LNG-IUS for cycle regulation and pain relief.
 - **Secondary causes:** Evaluate for endometriosis, adenomyosis, or pelvic pathology if symptoms persist.
 - **Adjuncts:** Heat therapy, exercise, lifestyle counseling.



Management of Menstrual Disorders (Updated ACOG 2024)...

- **Abnormal Uterine Bleeding (AUB)**
 - **Classification:** Use **PALM–COEIN system** (Polyp, Adenomyosis, Leiomyoma, Malignancy; Coagulopathy, Ovulatory dysfunction, Endometrial, Iatrogenic, Not classified).
 - **Acute management:**
 - High-dose estrogen or combined OCPs.
 - Tranexamic acid or NSAIDs.
 - **Chronic management:**
 - LNG-IUS (most effective long-term option).
 - Cyclic progestins or combined OCPs.
 - Surgical: hysteroscopic polypectomy, myomectomy, endometrial ablation, hysterectomy depending on etiology and fertility goals.
 - **Adolescents:** Focus on medical therapy and avoid invasive procedures unless necessary.
 - **Perimenopausal women:** Rule out malignancy with biopsy before initiating therapy.



Management of Menstrual Disorders (Updated ACOG 2024)...

- **Menorrhagia (Heavy Menstrual Bleeding)**
 - **Medical therapy:**
 - NSAIDs (reduce blood loss by 20–50%).
 - Tranexamic acid (antifibrinolytic, effective for acute heavy bleeding).
 - Hormonal therapy: combined OCPs, cyclic progestins, LNG-IUS (highly effective in reducing blood loss).
 - **Iron supplementation:** Essential to correct anemia.
 - **Surgical options:** Endometrial ablation, myomectomy, or hysterectomy if medical therapy fails and fertility is not desired.
 - **Underlying causes:** Treat fibroids, adenomyosis, or coagulopathies.



Management of Menstrual Disorders (Updated ACOG 2024)...

- **Resource-Limited Settings (ACOG Adaptation)**
 - **Clinical diagnosis prioritized** when imaging or labs are unavailable.
 - **Affordable therapies:** NSAIDs, oral iron, and combined OCPs are emphasized.
 - **Urine pregnancy test** is essential to exclude pregnancy-related bleeding.
 - **Referral pathways:** Establish clear criteria for surgical referral (e.g., persistent anemia, suspected malignancy).
 - **Education & counseling:** Key to adherence and early recognition of complications.



Hormone Therapy in Women – Key Concepts

Used for menopausal symptoms: hot flushes, urogenital atrophy, sleep disturbances.

Estrogen alone for women with hysterectomy; estrogen + progestogen otherwise.

Transdermal estrogen preferred in women at increased VTE risk.



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Why Estrogen alone for
women with
hysterectomy; not
estrogen + progestogen ?



Evidence from Previous Studies

Estrogen most effective for vasomotor symptoms (evidence-dependent efficacy).

Progestogen addition prevents endometrial hyperplasia.

Alternatives: SSRIs, venlafaxine, clonidine, gabapentin.

Contraindications include breast cancer, thromboembolism, liver disease.



Updated Gynecology Guidelines (ACOG/NAMS 2023–2024)

HT individualized: lowest effective dose, shortest duration.

Start HT ideally within 10 years of menopause for safer cardiovascular outcomes.

Avoid oral estrogen in patients with high triglycerides or gallbladder disease.

Non-hormonal therapy preferred in high-risk breast cancer patients.



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who are high-risk
breast cancer
patients?



Risks & Safety Considerations

Increased risk: VTE (particularly first year), stroke (dose-related).

Breast cancer risk increases with combined estrogen–progestogen after 3–5 years.

Transdermal estrogen lowers risk of VTE and gallbladder disease.



Premature Ovarian Insufficiency

Requires HT until natural age of menopause to protect bone and cardiovascular health.

Low-dose estrogen initiated and titrated upward.



Practical Takeaway

**NSAIDs, hormonal therapy,
and LNG-IUS
(Levonorgestrel-Releasing
Intrauterine System)**
remain the backbone of
management.

Tranexamic acid is
highlighted for acute heavy
bleeding.

**PALM-COEIN
classification** ensures
systematic evaluation.

Shared decision-making
and tailoring to patient
context are central to
ACOG 2024
recommendations.



References

- Pharmacotherapy Handbook, 7th Edition – Chapter 31.
- ACOG Practice Bulletin: Management of Menopausal Symptoms, 2023.
- North American Menopause Society (NAMS) Guidelines, 2023.
- ACOG Bulletin on Abnormal Uterine Bleeding, 2024.

